

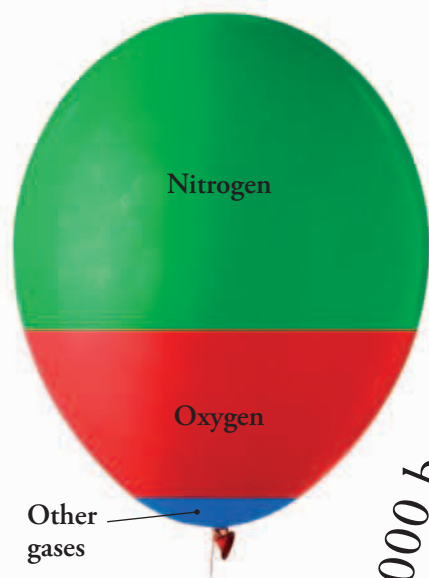
In and out

Your body cells need constant supplies of oxygen. They also need to get rid of waste carbon dioxide gas. This exchange of gases happens in your lungs. Breathing in and out ensures that fresh supplies of oxygen are brought in and stale air containing carbon dioxide is removed. The lungs cannot move by themselves, so the process relies on muscles in the chest to suck air in and push it out.

AIR IN

Air is drawn into the body by the movement of the rib muscles and the diaphragm—a sheet of muscles under the lungs. When you breathe in, the diaphragm tightens and pulls down. At the same time, your rib muscles tighten and pull your ribcage upward and outward.

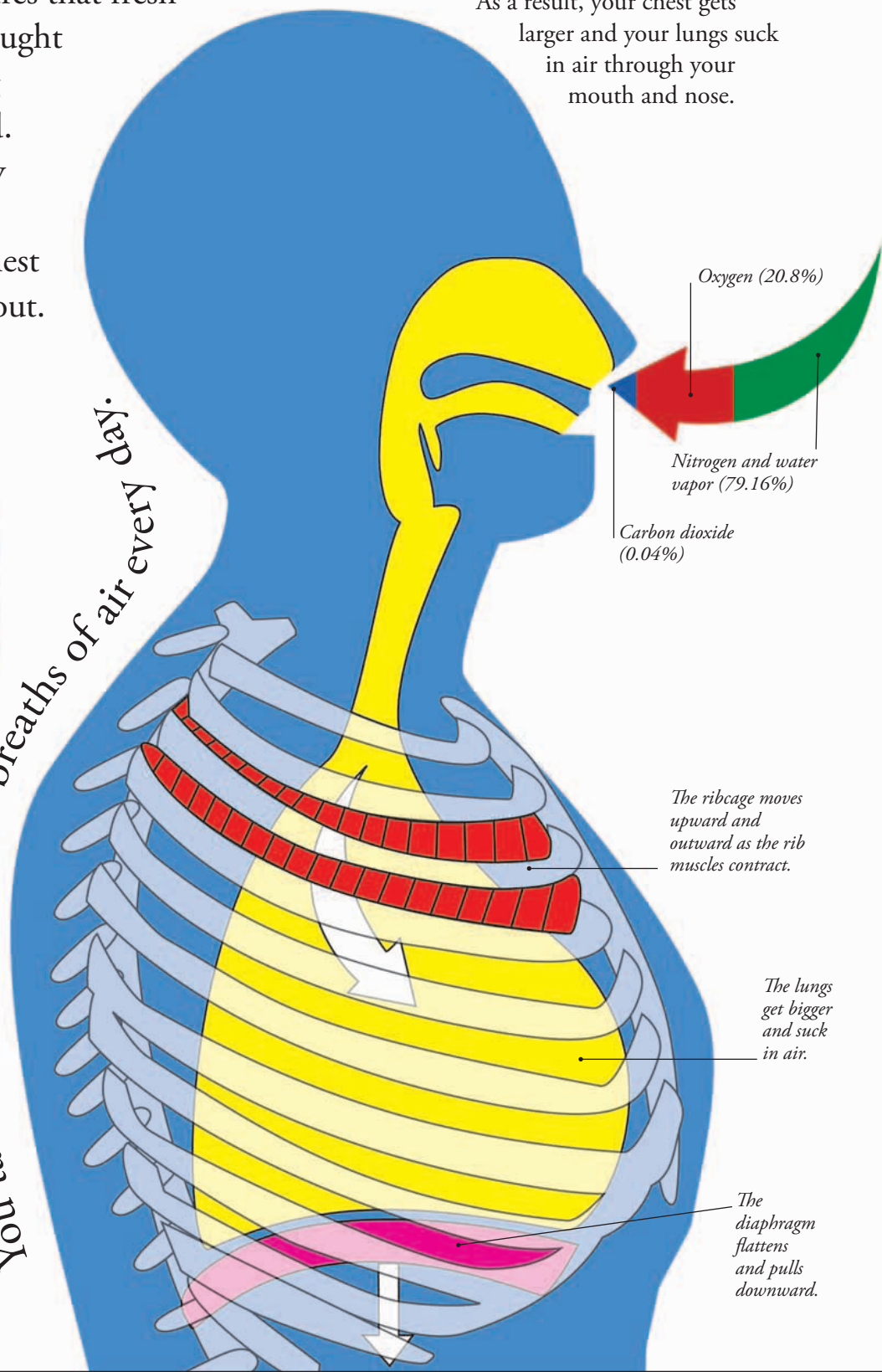
As a result, your chest gets larger and your lungs suck in air through your mouth and nose.



WHAT IS AIR?

Air is a mixture of gases that surrounds our planet. As you can see from this balloon diagram, air is mostly nitrogen (about 78 percent) and oxygen (about 21 percent), plus small amounts of other gases. Oxygen is the only part of the gas mixture that your body uses.

You take in about 40,000 breaths of air every day.



The ribcage moves upward and outward as the rib muscles contract.

The lungs get bigger and suck in air.

The diaphragm flattens and pulls downward.